Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, SQL, R
Data Analysis: Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing, Statistical Analysis, Frequency/Recency Analysis, Data Validation
SQL & Databases: SQL Queries, Joins, Aggregations, SQLite, MySQL, Database Design, Data Modeling, Data Quality Checks
Business Intelligence & Visualization: Power BI, DAX, Dashboard Development, Data Visualization, KPI Reporting, Customer Segmentation, Revenue Analysis
Data Science & Machine Learning: Scikit-learn, Predictive Modeling, BG/NBD, Gamma-Gamma Models, Anomaly Detection, Feature Engineering, Model Evaluation
Automation & Tools: Python Scripting, ETL Pipelines, Git, GitHub, Jupyter Notebook

EXPERIENCE

AI for Sustainability Virtual Intern* |LINK** — ***July – Sep, 2026

- Completed an AI-focused virtual internship covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

***Customer Lifetime Value Engine* |LINK**

- Built BG/NBD and Gamma-Gamma probabilistic models on 1M+ transaction records to predict customer lifetime value and purchasing behaviour, incorporating frequency/recency analysis and model validation.
- Performed transaction-level preprocessing and value-based customer segmentation, identifying the top 20% of customers contributing ~65% of forecasted revenue.
- Developed Power BI dashboards to visualise customer segments, revenue distribution, and lifetime value predictions, enabling data-driven business decisions and customer prioritisation.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered 15+ time-series features from 500K+ NSE market observations and trained Isolation Forest and LSTM Autoencoder models to detect anomalous price and volume movements.
- Applied FinBERT sentiment analysis to 25K+ financial news records and integrated sentiment scores with market features, combining structured market data and unstructured financial text into a unified analytical pipeline.
- Benchmarked 2 anomaly-detection models using precision, recall, F1-score, and inference time, with the best model targeting ~86% precision, ~81% recall, and ~83% F1-score on the evaluation dataset.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built an OCR-based data extraction pipeline using EasyOCR, OpenCV, and NLP to transform unstructured receipts into structured transaction data, improving processing efficiency by 40%.
- Developed data preprocessing and expense categorization workflows to clean, validate, and organize extracted transaction data for downstream analysis.
- Built analytics and reporting features for spend trends and budget tracking, converting transaction data into actionable financial insights for end users.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025